Solutions to JEE Main - 14 | JEE-2022

PHYSICS

SECTION-1

1.(C) Impedance, $Z = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2} = 1250\ \Omega$

Solving, we get $R = 750\ \Omega$

So, at angular frequency 400 rad/s,

$$Z = \sqrt{(750)^2 + \left(\frac{1}{(400)\left(5\times10^{-6}\right)}\right)^2} = \sqrt{(750)^2 + (500)^2} = 250\sqrt{13} \approx 901.4\ \Omega$$

2.(B) In the situation of minimum deviation, the ray travelling inside the prism is parallel to its base, and hence the angle of refraction is $\frac{A}{2} = 30^\circ$

By Snell's law, $\mu_{prism} = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2}$

3.(B) Let the amplitude of the voltage source be V

Rate of heat dissipation is given by $\frac{dQ}{dt} = \frac{V^2 R}{2Z^2} = \frac{V^2 R}{2\left(R^2 + (\omega L)^2\right)}$

So, if the frequency is increases, $\frac{dQ}{dt}$ decreases continuously

4.(D) We know that if the voltage source is $V(t) = V_0 \cos(\omega t)$, then the current is given by

$$I(t) = \frac{V_0}{Z}\cos(\omega t \pm \phi)$$

So, by comparing the given equations, we get $\frac{100}{Z} = 2 \quad \Rightarrow \quad Z = 50\ \Omega$

Also, $\phi = 60^\circ$ Now, since $\frac{R}{Z} = \cos\phi$, $R = (50)\cos 60^\circ = 25\,\Omega$

5.(D) Fringe width, $\beta = \frac{\lambda D}{d} = \frac{\left(500\times10^{-9}\right)(4)}{10^{-4}} = 2\text{ cm}$

6.(A) We know that the combination of a lens and mirror behaves like a mirror, and hence we must apply mirror equations

The focal length of this mirror is given by

$$\frac{1}{F} = \frac{-2}{40} + \frac{1}{\infty} \qquad F = -20\text{ cm}$$

We are given that the magnification is –2

$$\Rightarrow \quad -\frac{v}{u} = -2 \quad \Rightarrow \quad v = 2u$$

Mirror equation: $\frac{1}{2u} + \frac{1}{u} = \frac{1}{-20} \quad \Rightarrow \quad u = -30\text{ cm}$

7.(C) At a point at a distance y from the central maximum, the intensity is

$$I = I_M \cos^2\left(\frac{\Delta\phi}{2}\right) = I_M \cos^2\left(\frac{\pi y}{\beta}\right)$$

So, for $y = 0,\ I = I_M$ and for $y = \frac{\beta}{6}, \frac{5\beta}{6},\ I = \frac{3I_M}{4}$

So, for the minimum distance between any two such points, we must choose $y = \frac{\beta}{6}$ and $y = -\frac{\beta}{6}$

Therefore, the distance is $\Delta y = \frac{\beta}{6} - \left(-\frac{\beta}{6}\right) = \frac{\beta}{3}$

8.(D) Mirror formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{-60}$ $\quad v = 15\text{ cm}$

Magnification, $M = -\frac{v}{u} = \frac{1}{4}$

9.(B) The amplitudes of V_C, V_L and I can be anything as they depend on the values of R, L and C

But the phase relationship between V_C, V_L and I is always the same:

V_C must be behind I by a phase angle of $\frac{\pi}{2}$, and V_L must be ahead of I by a phase angle of $\frac{\pi}{2}$

10.(C) Because the incidence on face AB is normal, the ray suffers no deviation at face AB and strikes the face AC at an angle of incidence $30^o\ (= \angle A)$

Now, the ray suffers TIR at face AC if

$\sin 30^o > \frac{1}{\mu}$ i.e. if $\mu > 2$

Therefore, in both the given cases, TIR does not take place and the ray emerges into air from face AC itself

For $\mu = \sqrt{3}$, the angle of emergence e is given by

$$\frac{\sin 30^o}{\sin e} = \frac{1}{\sqrt{3}} \quad \Rightarrow \quad e = 60^o$$

The deviation suffered in any refraction is $|i - r|$

Since no deviation is suffered by the ray in crossing face AB, the net deviation is equal to the deviation suffered at face AC, i.e. $\delta = 60^o - 30^o = 30^o$

11.(A) Lens Makers' Formula: $\frac{1}{f} = \left(\frac{1.5}{4/3} - 1\right)\left(\frac{1}{-30} - \frac{1}{60}\right) = \left(\frac{1}{8}\right)\left(-\frac{1}{20}\right) \Rightarrow f = -160\text{ cm}$

12.(C) We know that $\delta = 180^o - 2\theta \Rightarrow \theta = 60^o$

13.(B) At the point where the third order minimum is formed, the phase difference is $\left(2(3) - 1\right)\frac{\lambda}{2} = \frac{5\lambda}{2}$

At a point on the screen located at a distance y from the central maxima, the path difference,

$$\Delta P = \left(\frac{yd}{D}\right) \quad \text{So,} \quad y = \left(\frac{D}{d}\right)\Delta P = \left(\frac{2}{0.2 \times 10^{-3}}\right)\left(\frac{5\left(500 \times 10^{-9}\right)}{2}\right) = 1.25\text{ cm}$$

14.(D) First refraction: $\frac{\mu}{v_1} - \frac{1}{-\infty} = \frac{\mu - 1}{+R} \quad \Rightarrow \quad v_1 = \left(\frac{\mu}{\mu - 1}\right)R$

Object distance for the second refraction, $u_2 = v_1 - 2R = \left(\frac{2-\mu}{\mu-1}\right)R$

So, for the second refraction:

$$\frac{1}{v_2} - \frac{\mu}{\left(\frac{2-\mu}{\mu-1}\right)R} = \frac{1-\mu}{-R} \quad \Rightarrow \quad \frac{1}{v_2} = \frac{\mu(\mu-1)}{(2-\mu)R} + \frac{\mu-1}{R} \quad \Rightarrow \quad v_2 = \frac{(2-\mu)}{2(\mu-1)}R$$

So, the distance of the final image from the centre of the sphere is $D = v_2 + R = \frac{\mu}{2(\mu-1)}R$

15.(C) Angular fringe width $= \frac{\beta}{D} = \frac{\lambda}{d}$ So, $\lambda = \left(5 \times 10^{-4}\right)\left(1.2 \times 10^{-3}\right) = 600$ nm

16.(B) We know that $I_{MAX} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = \left(\sqrt{I} + \sqrt{\alpha I}\right)^2 = \left(1 + \sqrt{\alpha}\right)^2 I$

And, $I_{MIN} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = \left(\sqrt{I} - \sqrt{\alpha I}\right)^2 = \left(1 - \sqrt{\alpha}\right)^2 I$

So, $\frac{I_{MAX}}{I_{MIN}} = \left(\frac{1+\sqrt{\alpha}}{1-\sqrt{\alpha}}\right)^2$

17.(B) Since the final image is the same size as the object, and the net magnification produced by the lenses is $M_1 M_2$ (if M_1 and M_2 are the magnification produced by the individual lenses),

$$M_1 M_2 = 1 \quad \Rightarrow \quad M_2 = \frac{1}{M_1}$$

We know that the magnification by a lens becomes its reciprocal if the object distance and image distance are interchanged. So, since the focal length of the lenses is the same, it is clear that if the image distance for lens 1 is *X*, then the object distance for lens 2 will be *X* too. And since the sum of these quantities, i.e. 2*X*, must be equal to the distance between the lenses, we get $2X = 120$ cm,

i.e. $X = 60$ cm

So, the image distance for the first lens is 60 cm

Applying the lens formula on the first lens, $\frac{1}{+60} - \frac{1}{u} = \frac{1}{45} \quad \Rightarrow u = -180$ cm

18.(A) Power factor at frequency $f = \frac{R}{Z_f}$

(where $Z_f = \left(R^2 + \left(\frac{1}{2\pi f C} - 2\pi f L\right)^2\right)^{1/2}$ is the impedance at frequency *f*)

Power consumed at frequency *f*, $P = \frac{V^2 R}{2Z_f^2}$

At resonance, the impedance is $Z_0 = R$, so the power consumed is

$$P_0 = \frac{V^2 R}{2R^2} = \frac{V^2}{2R}$$

So, $\frac{P}{P_0} = \frac{R^2}{Z_f^2} \quad \Rightarrow \quad \frac{R}{Z_f} = \left(\frac{P}{P_0}\right)^{1/2}$

19.(A) Convex lens: $\frac{1}{f_1}=\left(\frac{\mu_1}{1}-1\right)\left(\frac{1}{R_1}-\frac{1}{\infty}\right)$

Concave lens: $\frac{1}{f_2}=\left(\frac{\mu_2}{1}-1\right)\left(\frac{1}{-R_2}-\frac{1}{R_2}\right)$

Combination: $\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{(\mu_1-1)}{R_1}-\frac{2(\mu_2-1)}{R_2}$

The combination behaves like a converging lens if the equivalent focal length is positive

So, $\frac{(\mu_1-1)}{R_1}>\frac{2(\mu_2-1)}{R_2}$

i.e. $\frac{R_1}{(\mu_1-1)}<\frac{R_2}{2(\mu_2-1)}$

20.(B) Lens Formula: $\frac{1}{v}=\frac{1}{f}+\frac{1}{u}=\frac{1}{f}+\frac{1}{-(2f+\delta)}$

$\Rightarrow \quad v=\frac{(2f+\delta)f}{f+\delta}$

So, the magnification is $M=\frac{v}{u}=\frac{f}{f+\delta}=\left(1+\frac{\delta}{f}\right)^{-1}\approx 1-\frac{\delta}{f}$

SECTION-2

21.(0.01) Quality factor, $Q=\frac{1}{R}\sqrt{\frac{L}{C}}=\frac{1}{5\times10^3}\sqrt{\frac{20\times10^{-3}}{8\times10^{-6}}}=0.01$

22.(2) Shift on screen due to introduction of sheet, $S=\frac{Dt(\mu-1)}{d}=\frac{(1)(2\times10^{-4})(1.5-1)}{5\times10^{-3}}=2\text{ cm}$

23.(400) We know that $I_{RMS}=\frac{V}{\sqrt{2}Z}$

Since the amplitude V of the voltage source remains the same, $\frac{Z_2}{Z_1}=\frac{I_{RMS,1}}{I_{RMS,2}}=\frac{3}{4}$

$$\Rightarrow \quad 9\left(6^2+((200)(0.04))^2\right)=16\left(6^2+\left(\frac{1}{(200)C}-(200)(0.04)\right)^2\right)$$

$$\Rightarrow \quad \frac{1}{(200)C}=12.5 \quad \Rightarrow C=\frac{1}{(200)(12.5)}=400\ \mu\text{F}$$

24.(80) The water "lens" and the mirror behave together like a concave mirror

If the focal length of the original concave mirror is $-X$, its radius of curvature is $2X$

Now, the focal length of the water "lens" is given by

$$\frac{1}{f_{WL}} = \left(\frac{4/3}{1} - 1\right)\left(\frac{1}{\infty} - \frac{1}{-(2X)}\right) = \frac{1}{6X}$$

So, the equivalent focal length of the combination of the water "lens" and the original concave mirror is given by

$$\frac{1}{F} = \frac{-2}{6X} + \frac{1}{-X} \quad \Rightarrow \quad F = -\frac{3X}{4}$$

For the image to be coincident with the object, the object should be at the centre of curvature of the combination mirror

So, $\quad 2|F| = 120 \quad \Rightarrow \quad X = 80 \text{ cm}$

25.(9) When the experiment is performed inside water, the wavelength of the light becomes $\frac{\lambda}{n} = \frac{\lambda}{4/3}$

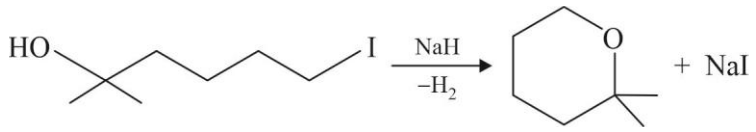
Fringe width of the first experiment, $\beta_1 = \frac{\lambda D}{d} = 6 \text{ mm}$

Therefore, the fringe width of the second experiment, $\beta_2 = \dfrac{\left(\dfrac{\lambda}{4/3}\right)(2D)}{d} = \dfrac{3}{2}\beta_1 = 9 \text{ mm}$

Chemistry

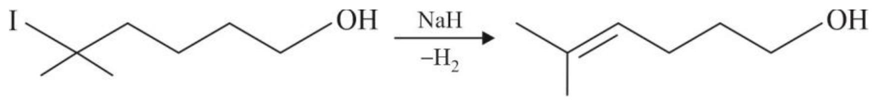
SECTION-1

1.(D) **I.**

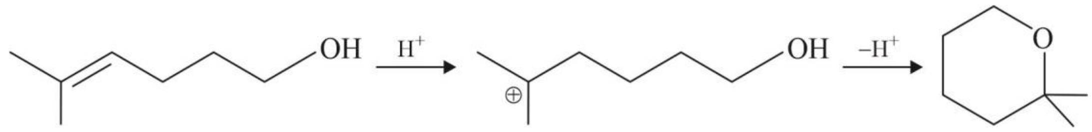


Intramolecular Williamson's ether synthesis

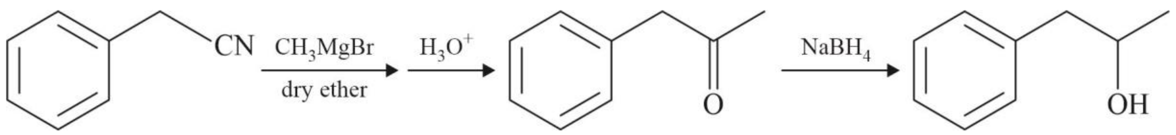
II.



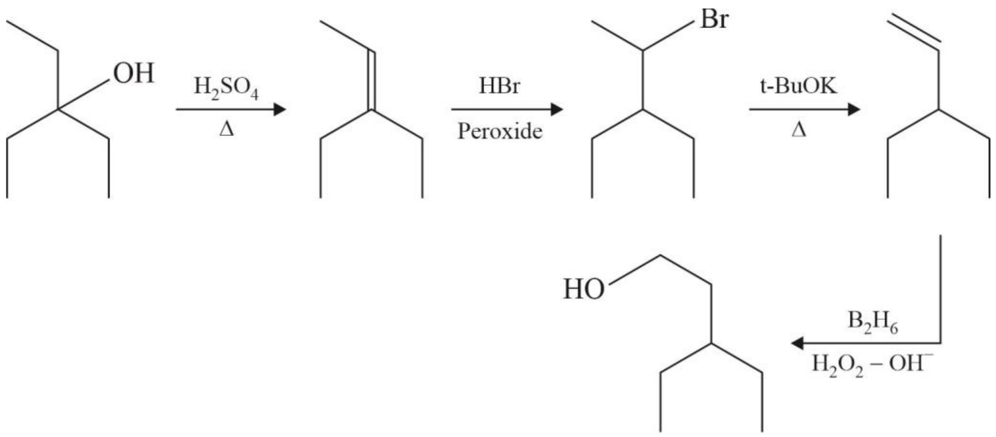
Intramolecular E2



2.(C)

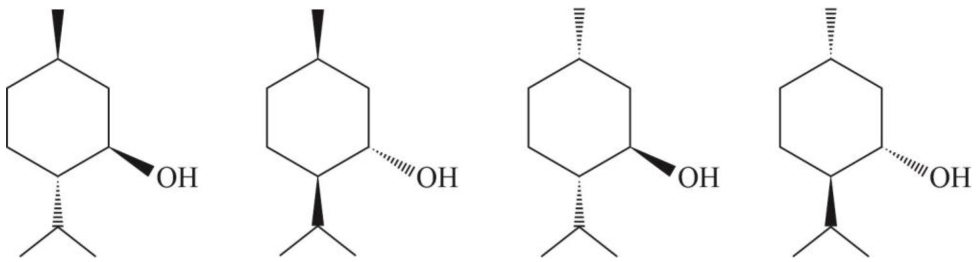


3.(C)



4.(B)

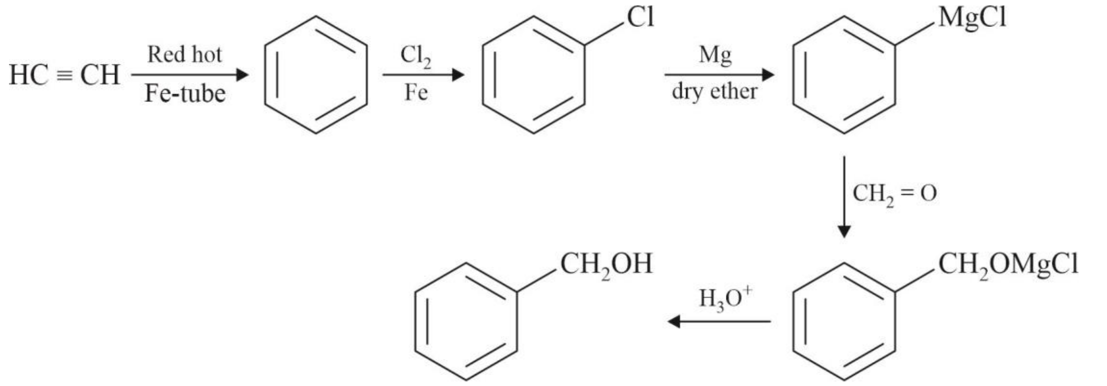
5.(B)



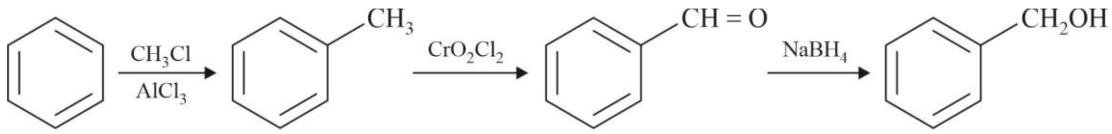
6.(C) Light corresponding to the energy of red region is absorbed by the complex.

7.(B) Wilkinson's catalyst is $[Rh(PPh_3)_3Cl]$.

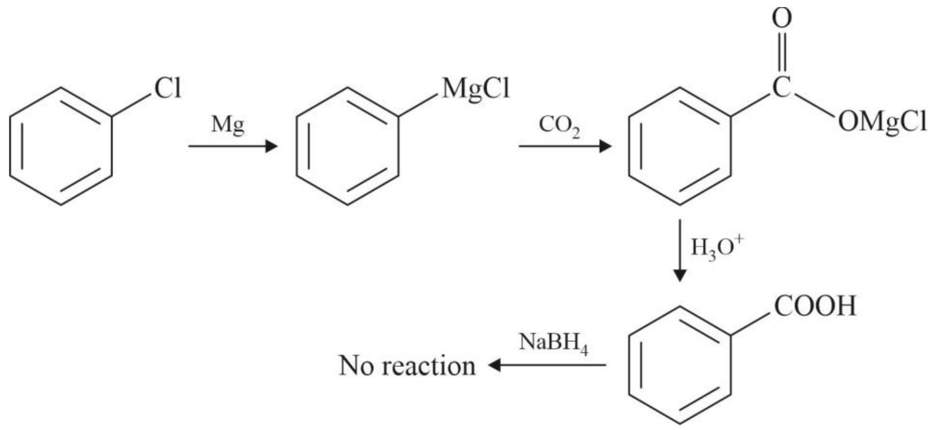
8.(C) **I.**



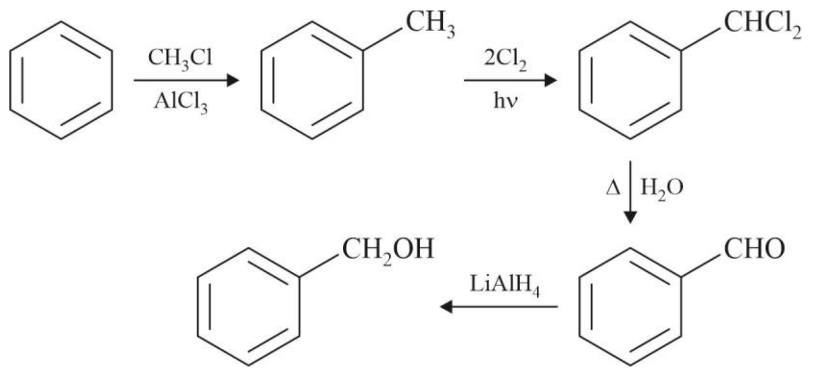
II.



III.

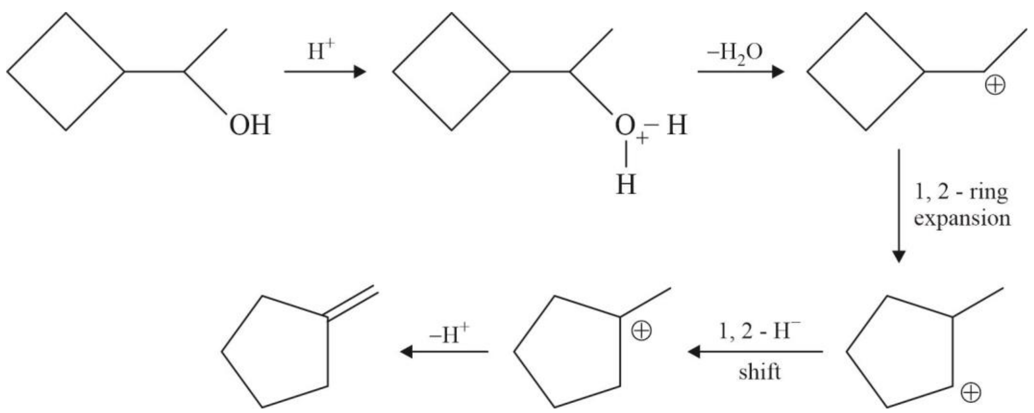


IV.

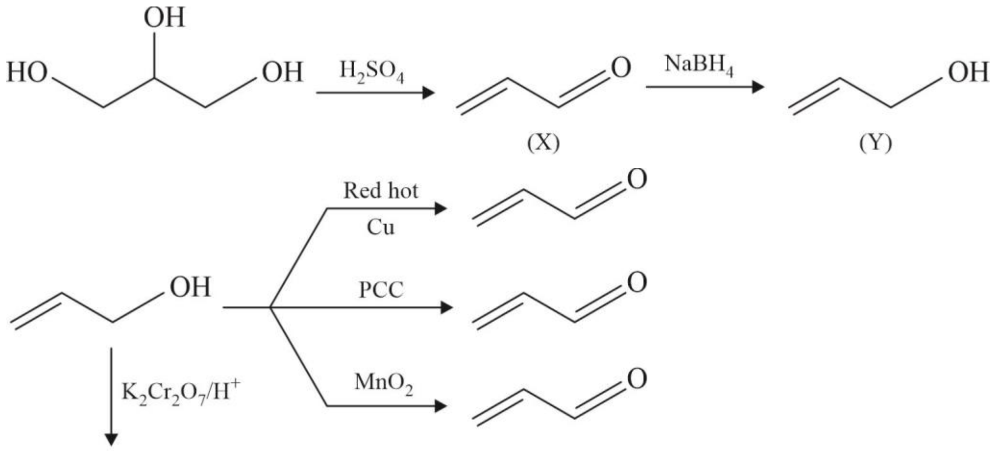


9.(B) It is diamagnetic complex

10.(B)

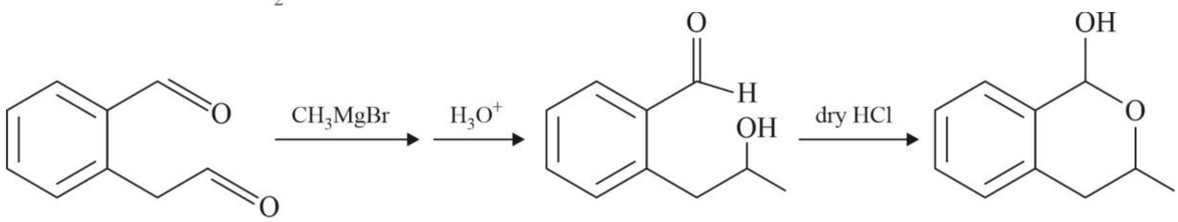


11.(D)



$HOOC-COOH+CO_2$

12.(A)

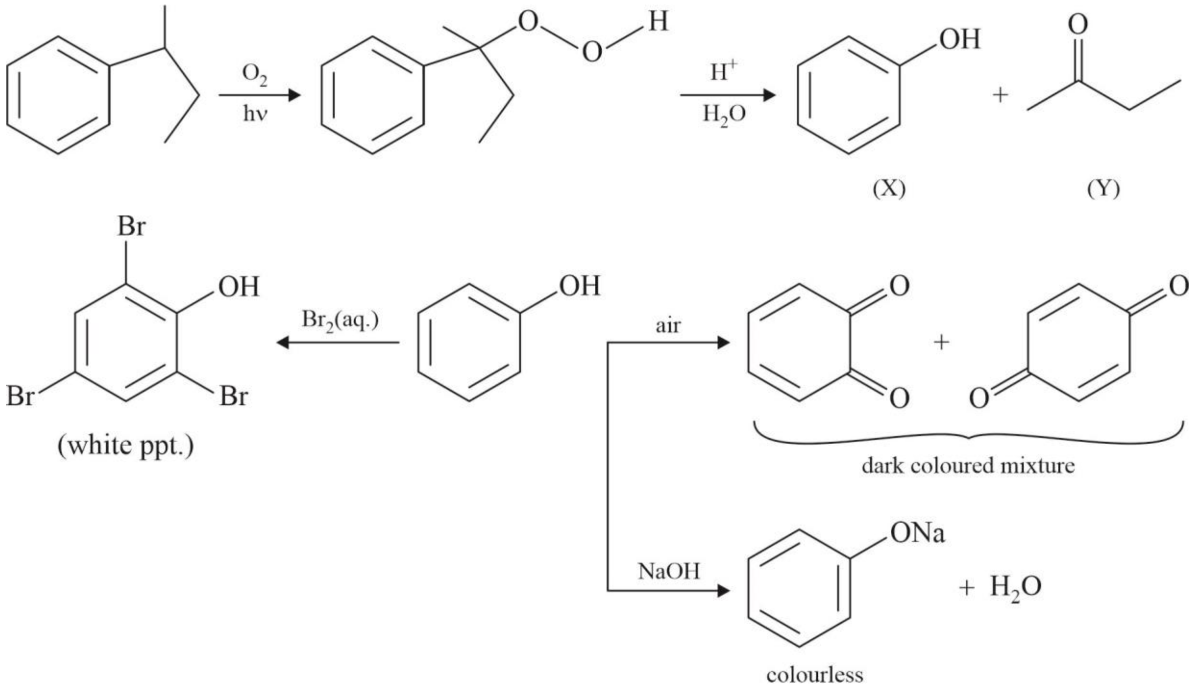


13.(C) $B_2H_6+6ROH \longrightarrow 2B(OR)_3+6H_2$

Isomer of THF is cyclobutanol.

14.(D) Picric acid is prepared by treating phenol first with concentrated H_2SO_4 which converts it to phenol-2, 4-disulphonic acid, and then with conc. HNO_3 to get 2, 4, 6-trinitrophenol.

15.(C)



16.(D) Most of the methanol is produced by catalytic hydrogenation of carbon monoxide at high pressure and temperature in the presence of $ZnO-Cr_2O_3$ catalyst.

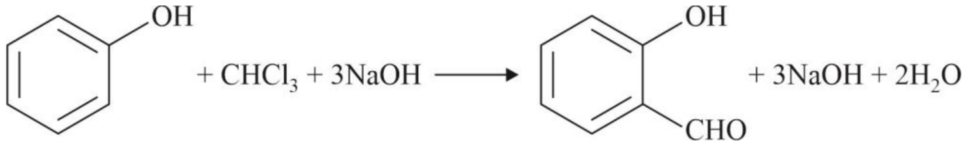
17.(C)

$Co(C_2O_4)Br \cdot 4NH_3$

$[Co(C_2O_4)(NH_3)_4]Br$
(X)
Cationic complex

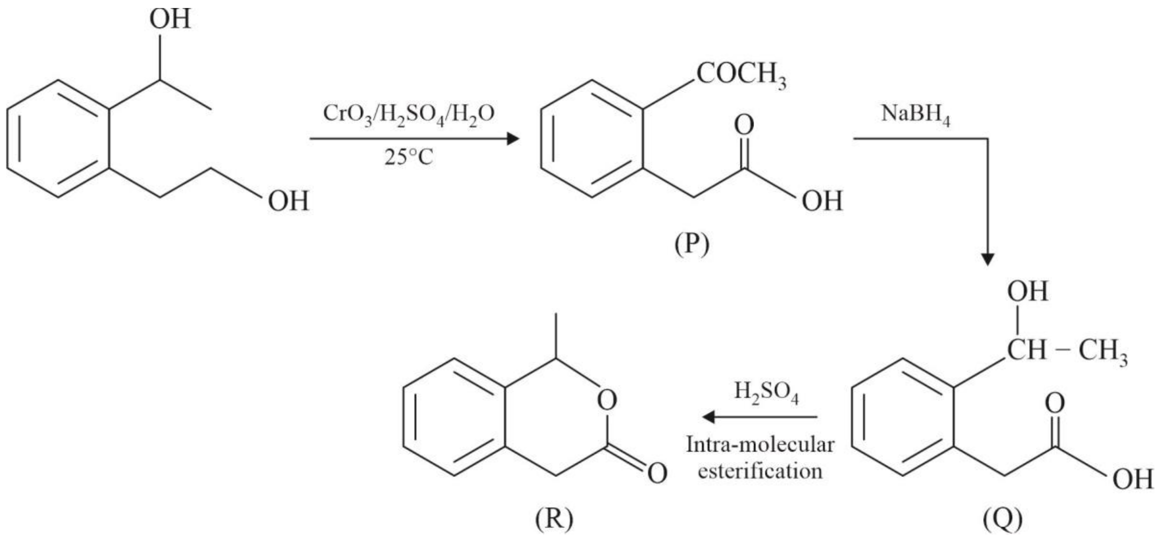
$(NH_4)[Co(C_2O_4)(NH_2)Br(NH_3)_2]$
(Y)
Anionic complex

18.(A)



19.(A) Metal carbonyl shows synergic bonding

20.(D)



SECTION-2

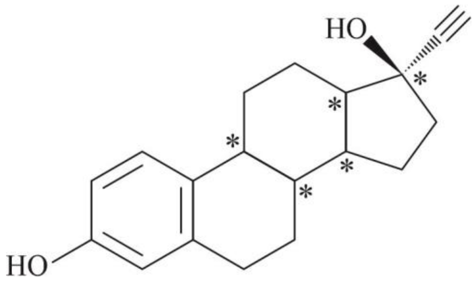
21.(90) $(NH_4)_2[Ce(NO_3)_6]$: Here NO_3^- is bidentate ligand.
$EAN = (58-4)+24 = 78$

22.(6)

(a)	Neither	**(b)**	Neither	**(c)**	Reduction	**(d)**	Neither
(e)	Neither	**(f)**	Neither	**(g)**	Oxidation	**(h)**	Oxidation
(i)	Oxidation	**(j)**	Neither				

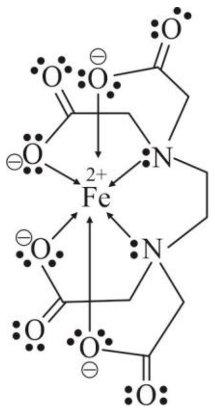
23.(3) Average bond order for $C-O$ bond $= \frac{3}{2}$; Aromatic having 2π - electrons $(n = 0)$

24.(19) $X_1 = 5;\ X_2 = 5;\ X_3 = 9$



$*$ $\Rightarrow$ Chiral carbon atoms
$*$ $\Rightarrow$ Saturated tertiary carbon atoms

25.(27) $Z_1 = 5;\ Z_2 = 16;\ Z_3 = 6$



Mathematics

SECTION-1

1.(D) $(\vec{a}\times\vec{b})\times\vec{V}-\vec{u}\times(\vec{d}\times\vec{a})=[\vec{a}\ \vec{c}\ \vec{d}]\vec{b}$

$(\vec{a}\cdot\vec{V})\vec{b}-(\vec{b}\cdot\vec{V})\vec{a}-(\vec{u}\cdot\vec{a})\vec{d}+(\vec{u}\cdot\vec{d})\vec{a}=[\vec{a}\ \vec{c}\ \vec{d}]\vec{b}$

$[\vec{a}\ \vec{c}\ \vec{d}]\vec{b}-[\vec{b}\ \vec{c}\ \vec{d}]\vec{a}-[\vec{b}\ \vec{c}\ \vec{a}]\vec{d}+[\vec{b}\ \vec{c}\ \vec{d}]\vec{a}=[\vec{a}\ \vec{c}\ \vec{d}]\vec{b}$

$[\vec{b}\ \vec{c}\ \vec{a}]\vec{d}=0 \quad \Rightarrow \vec{a},\vec{b},\vec{c}$ are coplanar

2.(A) Vector perpendicular to $2\hat{i}+\hat{j}-\hat{k}$ and $\hat{i}+3\hat{j}-\hat{k}$ is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 1 & 3 & -1\end{vmatrix}=2\hat{i}+\hat{j}+5\hat{k}$$

Any general point on the line is $(1+2\lambda, 1+\lambda, 1-\lambda)$ at their point of intersection. This point satisfies equation of plane.

$(1+2\lambda)+3(1+\lambda)-1(1-\lambda)=9 \Rightarrow \lambda=1 \qquad \therefore \qquad$ Point of intersection is (3, 2, 0).

Hence required line is $\vec{r}=(3\hat{i}+2\hat{j})+k(2\hat{i}+\hat{j}+5\hat{k}) \quad \Rightarrow \quad \dfrac{x-3}{2}=\dfrac{y-2}{1}=\dfrac{z}{5}$

3.(B) $M\equiv(4,5,6)$

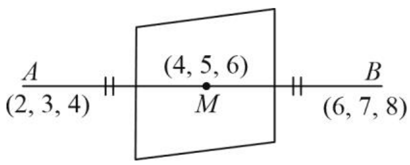
Plane passes through 4, 5, 6

$\therefore \quad A(x-4)+B(y-5)+C(z-5)=0$

A, B, C are 4, 4,4

$\therefore \quad$ equation is $x-4+y-5+z-6=0$

$\Rightarrow \quad x+y+z=15$



4.(C) L_1 through Point $A(1,1,1)$ and dir $<1,1,1>$

L_2 through point $B(1,-1,0)$ and dir $<1,-1,-1>$

$$\therefore \quad \text{S.D}=\frac{\begin{vmatrix}0 & 2 & 1\\ 1 & 1 & 1\\ 1 & -1 & -1\end{vmatrix}}{\sqrt{4+4+0}}=\frac{2}{2\sqrt{2}}=\frac{1}{\sqrt{2}}$$

5.(A) Line joins $P(3,-4,1)$ & $Q(2\lambda+3,-3\lambda-1,2-\lambda)$ and $\overrightarrow{PQ}\cdot\vec{n}=0$

$\Rightarrow \quad <2\lambda,-3\lambda+3,1-\lambda>\perp<2,1,-1>$

$\Rightarrow \quad 4\lambda-3\lambda+3-1+\lambda=0 \Rightarrow 2\lambda+2=0$

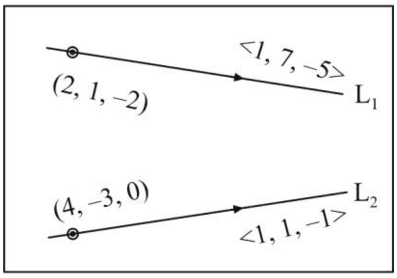
$\therefore \quad \lambda=-1$

$\therefore \quad P(3,-4,1)$ and $Q(1,2,3)$

$\therefore \quad \dfrac{x-3}{1-3}=\dfrac{y+4}{2+4}=\dfrac{z-1}{3-1} \Rightarrow \dfrac{x-3}{-2}=\dfrac{y+4}{6}=\dfrac{z-1}{2}$

$\therefore \quad$ line is $\dfrac{x-3}{1}=\dfrac{y+4}{-3}=\dfrac{z-1}{-1}$

6.(B) π is $x+2y+3z+2=0$



$\therefore \quad P=\left|\frac{2}{\sqrt{1+4+9}}\right|=\frac{2}{\sqrt{14}}$

7.(D) L_1 and L_2 are intersecting lines

The position vector of their point of intersection is $5\hat{i}-7\hat{j}+6\hat{k}$ (For $\lambda=2$ or $\mu=1$)

Also, angle between L_1 and $L_2=\frac{70}{11\sqrt{42}}$

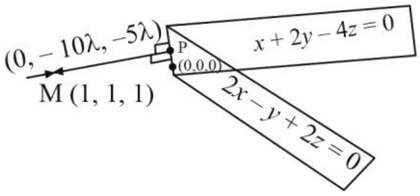
8.(A) Solving the equation of planes, we get equation of line containing planes

$$\frac{x}{0}=\frac{y}{-10}=\frac{z}{-5} \qquad \ldots(1)$$

Any point P on (1) is $(0,-10\lambda,-5\lambda)$.

Now, direction ratios of the line joining P and M is $<1, 1+10\lambda, 1+5\lambda>$

So, d.r's of MP are $\left\langle -1, \frac{1}{5}, \frac{-2}{5}\right\rangle$



So, equation of required line is $\frac{x-1}{5}=\frac{y-1}{-1}=\frac{z-1}{2}$

9.(A) z-axis is $\vec{r}=\vec{0}+\lambda(\hat{k})$

Line is $\vec{r}=(2\hat{i}+5\hat{j}-\hat{k})+\mu(3\hat{i}+2\hat{j}-5\hat{k})$

$\therefore$ shortest distance $=\left|\frac{(2\hat{i}+5\hat{j}-\hat{k})\cdot((3\hat{i}+2\hat{j}-5\hat{k})\times\hat{k})}{|(3\hat{i}+2\hat{j}-5\hat{k})\times\hat{k}|}\right|=\frac{11}{\sqrt{13}}$

10.(C) Equation of line parallel to given line through P is $\frac{x-1}{1}=\frac{y+5}{1}=\frac{z-9}{1}=\lambda$

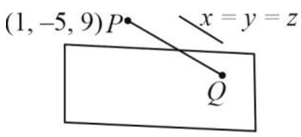
Its point of intersection with plane is $Q(\lambda+1, \lambda-5, \lambda+9)$

$\therefore \quad (\lambda+1)-(\lambda-5)+(\lambda+9)=5$

$\Rightarrow \quad \lambda=-10$

$\therefore \quad Q=(-9,-15,-1)$

$\therefore \quad PQ=10\sqrt{3}$



11.(D) $|\vec{r}\,[\vec{a}\,\vec{b}\,\vec{c}]| = |\vec{r}|\,[\vec{a}\,\vec{b}\,\vec{c}] = 2\times3\times6 = 36$

12.(D) $\vec{n}_1 = (4, -3, 4)$

$\vec{n}_2 = (3, -2, 1)$

$\vec{n}_1 \times \vec{n}_2 = (5, 8, 1)$

$(\vec{n}_1 \times \vec{n}_2)\cdot\vec{n} = 0$

$\Rightarrow \quad 10 - 8 + a = 0 \;\Rightarrow\; a = -2$

13.(D) Let the ratio $t:1$

$\Rightarrow$ point $P \equiv \left(\dfrac{3t-2}{t+1}, \dfrac{-5t+4}{t+1}, \dfrac{8t+7}{t+1}\right)$ lies on plane

$$\Rightarrow \quad \left(\frac{3t-2}{t+1}\right)1 - 2\left(\frac{-5t+4}{t+1}\right) + 3\left(\frac{8t+7}{t+1}\right) = 17 \;\Rightarrow\; t = \frac{3}{10}$$

14.(A) Area $= \dfrac{1}{2}\sqrt{\left(\dfrac{k^2}{lm}\right)^2 + \left(\dfrac{k^2}{mn}\right)^2 + \left(\dfrac{k^2}{nl}\right)^2} = \dfrac{k^2}{2lmn}$

$$\Rightarrow \quad \frac{l^2+m^2+n^2}{3} \ge (l^2m^2\,n^2)^{1/3}$$

$$\Rightarrow \quad \frac{1}{3} \ge (l\,m\,n)^{2/3} \;\Rightarrow\; \frac{1}{3\sqrt{3}} \ge l\,mn \;\Rightarrow\; \frac{1}{lmn} \ge 3\sqrt{3} \;\Rightarrow\; \frac{k^2}{2\,lmn} \ge \frac{3\sqrt{3}\,k^2}{2}$$

15.(D) $\begin{vmatrix} a-1 & -1 & 0 \\ 1 & 1 & 3 \\ 1 & 1 & 2 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & -1 & 0 \\ 0 & 1 & 3 \\ 0 & 1 & 2 \end{vmatrix} = 0$

$\Rightarrow a(3-2) = 0 \Rightarrow a = 0$

16.(C) $\vec{r} = (\vec{a}\times\vec{b})\sin x + (\vec{b}\times\vec{c})\cos y + 2(\vec{c}\times\vec{a})$

$\vec{r}\cdot(\vec{a}+\vec{b}+\vec{c}) = 0$

$\Rightarrow [\vec{a}\,\vec{b}\,\vec{c}](\sin x + \cos y + 2) = 0 \Rightarrow [\vec{a}\,\vec{b}\,\vec{c}] \ne 0 \;\Rightarrow\; \sin x + \cos y = -2$

This is possible only when $\sin x = -1$ and $\cos y = -1$

For $x^2 + y^2$ to be minimum $x = -\dfrac{\pi}{2}$ and $y = \pi$

$\Rightarrow$ minimum value of (x^2+y^2) is $= \dfrac{\pi^2}{4} + \pi^2 = \dfrac{5\pi^2}{4}$

17.(C) $\vec{b}\times\vec{d} = 0 \quad \Rightarrow \vec{b} = \lambda\vec{d}$

$\vec{a} = \vec{b} + \vec{c}$ & $\vec{c}\cdot\vec{d} = 0 \;\Rightarrow\; \vec{a}\cdot\vec{d} = \vec{b}\cdot\vec{d} + \vec{c}\cdot\vec{d}$

or $\quad \vec{a}\cdot\vec{d} = \vec{b}\cdot\vec{d}$

Now $\dfrac{\vec{d}\times(\vec{a}\times\vec{d})}{\vec{d}^2} = \dfrac{(\vec{d}\cdot\vec{d})\vec{a} - (\vec{d}\cdot\vec{a})\vec{d}}{\vec{d}^2} = \vec{a} - \dfrac{(\vec{b}\cdot\vec{d})\vec{d}}{\vec{d}^2} = \vec{a} - \dfrac{(\lambda\vec{d}\cdot\vec{d})\vec{d}}{d^2} = \vec{a} - \lambda\vec{d} = \vec{a} - \vec{b} = \vec{c}$

18.(A) The given planes can be written as $-2x+y-z+2=0$ and $-x-2y+z+3=0$

Here, $(-2)(-1)+(1)(-2)+(-1)(1)=-1<0$

Equation of bisectors $\dfrac{(-2x+y-z+2)}{\sqrt{(4+1+1)}}=\pm\dfrac{(-x-2y+z+3)}{\sqrt{1+4+1}}$

$\therefore$ Acute angle bisector is

$(-2x+y-z+2)=(-x-2y+z+3) \Rightarrow x-3y+2z+1=0$

19.(C) $\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times\vec{b})))))=\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times(0-4\vec{b}))))$

in similar manner $=(-4)^3\,\vec{b}=-64\vec{b}$

20.(A) Equation of plane $2x-3y+5z+7=0$

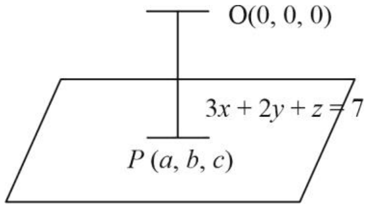
Equation of plane, parallel to the plane $2x-3y+5z+7=0$ is $2x-3y+5z=\lambda$

Pass through (3, 4, 1) then $6-12+5=\lambda \Rightarrow \lambda=-1$

$\therefore$ equation of plane $2x-3y+5z+1=0$

SECTION-2

21.(3.50) Clearly minimum value of $a^2+b^2+c^2$



$$=\left(\frac{|(3(0)+2(0)+(0)-7|}{\sqrt{(3)^2+(2)^2+(1)^2}}\right)^2=\frac{49}{14}=\frac{7}{2}\text{ units.}$$

(This is possible when $P(a,b,c)$ is foot of perpendicular from $O(0,0,0)$ on the plane.)

22.(6) We have $[\hat{a}\ \hat{b}\ \hat{a}\times\hat{b}]=\dfrac{1}{4}$

$\Rightarrow\quad (\hat{a}\times\hat{b})\cdot(\hat{a}\times\hat{b})=\dfrac{1}{4} \Rightarrow |\hat{a}\times\hat{b}|^2=\dfrac{1}{4}$

$\Rightarrow\quad \sin^2\theta=\dfrac{1}{4} \Rightarrow \sin\theta=\dfrac{1}{2}$

Hence $\theta=\dfrac{\pi}{6}$ (As $|\vec{a}|=1=|\vec{b}|$)

23.(58) Since, both the planes are parallel

$P_1: 4x-6y+12z+10=0$

$P_2: 4x-6y+12z+d=0$

$b=-6, c=12$

Now, $\left|\dfrac{d-10}{2\sqrt{4+9+36}}\right|=3$

$|d-10|=42 \;\Rightarrow\; d=52$ or -32

$\therefore\; P_2$ is $4x-6y+12z+52=0$

or $\quad 4x-6y+12z-32=0$

$\because$ Point $(-3, 0, -1)$ is lying between planes P_1 and P_2

$\therefore$ On substituting the point in the equation of the planes both expressions must be of opposite sign.

From $P_1: 4\times(-3)-6\times 0+12(-1)+10=-\text{ve}$

From $P_2: 4\times(-3)-6\times 0+12(-1)+52=+\text{ve}$

$\therefore\; d$ must be 52 $\qquad$ Hence, $(b+c+d)=-6+12+52=58$

24.(15) $\because \quad |\vec{c}-\vec{a}|=2\sqrt{2} \;\Rightarrow\; |\vec{c}-\vec{a}|^2=8$

$\Rightarrow \quad |\vec{c}|^2+9-2\,\vec{c}\cdot\vec{a}=8 \;\Rightarrow |\vec{c}|^2-2|\vec{c}|+1=0 \;\Rightarrow\; |\vec{c}|=1$

$$\Rightarrow \quad \vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -2\\ 1 & 1 & 0\end{vmatrix}=2\hat{i}-2\hat{j}+\hat{k}$$

$$\therefore \quad 10|(\vec{a}\times\vec{b})\times\vec{c}|=10|\vec{a}\times\vec{b}|\;|\vec{c}|\sin 30^\circ=10\times 3\times 1\times\frac{1}{2}=15$$

25.(8) $(x\hat{i}+y\hat{j})\cdot(x\hat{i}+y\hat{j}+8\hat{i}-10\hat{j})+41=0$

$x^2+y^2+8x-10y+41=0$

Centre (-4, 5), $r=\sqrt{16+25-41}=0$

So, minimum value of $|-4\hat{i}+5\hat{j}+2\hat{i}-3\hat{j}|^2=|-2\hat{i}+2\hat{j}|^2=(\sqrt{4+4})^2=8$